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I am currently a second-year PhD candidate in astrophysics at The University of Hong Kong . My primary research interest centers on extragalactic fast X-ray transients (EFXTs), gamma-ray bursts (GRBs) and their associated astrophysical phenomena. The focus of my recent research has been on investigating the diverse populations of EFXTs detected by the Einstein Probe (EP), including Type I (compact object merger origin), Type II (collapsar origin), and Type III (supermassive black hole and other origins) EFXTs. I have conducted detailed studies of several representative EFXTs, including EP240219a, the first EFXT detected by EP associated with an untriggered GRB; EP250704a, a minutes-long Type I EFXT associated with a short GRB; and EP250304a, a supernova-associated Type II EFXT originating from a jet-cocoon system. In addition to case studies, I am conducting population analyses of Type I EFXTs to constrain the remnant type of compact object mergers, as well as systematic studies of the evolution of early X-ray absorption in EFXTs to probe their local environments.

EDUCATION

2025 – PRESENT **PhD student in Astrophysics**
Department of Physics
The University of Hong Kong

2023 – 2025 **M.S. in Astronomy**
School of Astronomy and Space Science
Nanjing University

2019 – 2023 **B.S. in Physics**
School of Physics
Nanjing University

RESEARCH INTEREST

- Fast X-ray transients and their populations
- Progenitors and physical origins of fast X-ray transients
- Gamma-ray bursts and their associated phenomena
- Compact object mergers and their gravitational-wave and electromagnetic counterparts

RESEARCH EXPERIENCE

MARCH 2023 – PRESENT

Einstein Probe Mission
Transient Advocate

The Einstein Probe is a mission of the Chinese Academy of Sciences dedicated to time-domain high-energy astrophysics. I have led the studies of two interesting Type II fast X-ray transients, EP240219a/GRB 240219A (Yin et al., 2024, ApJL) and EP250404a/GRB 250404A (Yin et al., 2025, ApJL). As an active EP Transient Advocate, I involve in the discovery and analysis of many LEIA/EP-detected fast X-ray transients, e.g., GRB 230307A/LXT230307a (Sun et al., 2024, National Science Review, participant), EP240315a (Liu et al., 2025, Nature Astronomy, participant), EP250704a (Li et al., 2026, Science Bulletin, corresponding author), and EP260321a (Yuan et al., 2026, Science under review, participant).

NOV 2025 – PRESENT

The Hong Kong Institute for Astronomy and Astrophysics
Member

Created by Prof. Bing Zhang, The Hong Kong Institute for Astronomy and Astrophysics (HKIAA) is an international center of research excellence in astronomy and astrophysics and a platform for fostering communication and collaboration within the international

astronomical community. I collaborate with Prof. Bing Zhang on population studies of fast X-ray transients and the physics of compact object merger remnants.

JAN 2022 – SEP 2025

NJU GRB Group
Former Member

As a member of Professor Bin-Bin Zhang's GRB research team, I participated in several GRB and GW related research projects. Specifically, I conducted a GW-related research of the merger-driven GRB 211211A (Yin et al., 2023, ApJL), which is a peculiar long-duration burst with kilonova emission (Yang et al., 2022, Nature, participant). Moreover, I led a project to analyze MGF GRB 231115A and confirmed its origin as an extragalactic magnetar giant flare powered by a Comptonized fireball (Yin et al., 2024, ApJL).

DEC 2020 – SEP 2025

GRID Project
Former Member

Named Gamma-Ray Integrated Detectors (GRID), the mission forms a full-time all-sky one gamma-ray detection network that monitors the transient gamma-ray sky in the multi-messenger astronomy era. The student-led CubeSats developed by the NJU team have been successfully launched and operated. I co-led projects analyzing GRBs detected by GRID, including GRB 220408B (Zhang et al., 2023, RAA) and GRB 230812B (Wang et al., 2024, ApJ).

PROFESSIONAL EXPERIENCE/ACTIVITIES

2026 **Oral Talks,** COSPAR 2026, Florence, Italy

- **Oral Talk,** Asia-Pacific Regional IAU Meeting (ARPIM) 2026, Hong Kong, China
- **Oral Talk,** The Third Shenkuo Academic Symposium of the Chinese Astronomical Society, Zhuhai, China

2025 **LOC,** The First HKIAA Symposium, Hong Kong, China

- **Oral Talk,** Einstein Probe: Extragalactic Fast Transients and Multi-messenger Time-domain Astronomy (2025) Workshop, Beijing, China

2024 **Invited Talk,** GRID 2024 Summer Camp, Tsinghua University

- **Teaching Assistant,** Gamma-Ray Bursts: From Devices and Data to Science, Nanjing University
- **Oral Talk,** COSPAR 2024, Busan, Korea
- **Organizer/Volunteer/Oral Talk/Poster,** The Third Nanjing GRB Conference, Suzhou, China
- **Teaching Assistant,** Gravitational Wave and Related Astrophysics, Nanjing University

REVIEWER/REFEREE

- **Nature Astronomy**
- **The Astrophysical Journal**
- **Journal of High Energy Astrophysics**

SCHOLARSHIPS AND HONORS

2025 **Postgraduate Scholarship**
2024 **National Scholarship for Graduate Student**
(Most prestigious scholarship for graduate student in China)
2023, 2024 **First Class Academic Scholarship**
2023 **Outstanding Graduate**
2021, 2022 **The People's Scholarship of China**

PUBLICATIONS

NUMBER OF PAPERS	21
TOTAL CITATIONS	699
H-INDEX	10

An up-to-date paper list on ADS is available on this **link**.

First Author/Corresponding Author:

2026 **Minutes-long soft X-ray prompt emission from a compact object merger.** A. Li, C.-W. Wang, et al., Science Bulletin, doi: 10.1016/j.scib.2026.08.021

2025 **Spectral Hardening Reveals Afterglow Emergence in Long-Duration Fast X-ray Transients: A Case Study of GRB 250404A/EP250404a.** Y.-H. I. Yin, Y. Fang, et al., ApJL, doi: 10.3847/2041-8213/adf552

2024 **Triggering the Untriggered: The First Einstein Probe-Detected Gamma-Ray Burst GRB 240219A and Its Implications.** Y.-H. I. Yin, B.-B. Zhang, et al., ApJL, doi: 10.3847/2041-8213/ad8652

- **A Componized Fireball Bubble Fits the Second Extragalactic Magnetar Giant Flare GRB 231115A.** Y.-H. I. Yin, Z. J. Zhang, et al., ApJL, doi: 10.3847/2041-8213/ad2839

2023 **GRB 211211A-like Events and How Gravitational Waves May Tell Their Origins.** Y.-H. I. Yin, B.-B. Zhang, et al., ApJL, doi: 10.3847/2041-8213/acf04a

Selected Co-author:

2026 **X-rays breaking out of pre-explosion ejecta mark a supernova's first light.** W. Yuan, Q.-J Huang, et al., Science under review, arXiv: 2606.10014

- **Probing a new subclass of llGRB-SN transients: Insights from EP250304a and its associated supernova.** L. Cotter, A. Martin-Carrillo, et al., A&A, arXiv: 2606.06213
- **An extremely bright slow-rising afterglow from an off-axis jet in GRB 260310A.** Y.-H. Yang, R. Ricci, et al., ApJ, arXiv: 2605.23818

2025 **Evidence for a brief appearance of gamma-ray periodicity after a compact star merger.** R.-C. Chen, B.-B. Zhang, et al., Nature Astronomy, doi: 10.1038/s41550-025-02649-w

- **EP240801a/XRF 240801B: An X-ray Flash Detected by the Einstein Probe and Implications of its Multiband Afterglow** S.-Q. Jiang, D. Xu, et al., ApJL, doi: 10.3847/2041-8213/addebf
- **Soft X-ray prompt emission from a high-redshift gamma-ray burst EP240315a** Y. Liu, H. Sun, et al., Nature Astronomy, doi: 10.1038/s41550-024-02449-8
- **Bridging the Gap: GRB 230812B – A Three-Second Supernova-Associated Burst Detected by the GRID Mission**, C.-Y. Wang, Y.-H. I. Yin, et al., ApJ, doi: 10.3847/1538-4357/ada934

2024 **Magnetar emergence in a peculiar gamma-ray burst from a compact star merger.** H. Sun, C.-W. Wang, et al., National Science Review, doi: 10.1093/nsr/nwae401

- **Prospects for detecting neutron star-white dwarf mergers: early warnings from decihertz gravitational-wave observatories.** Y. Kang, C. Liu, et al. MNRAS, doi: 10.1093/mnras/stae340

2023 **Synchrotron Radiation Dominates the Extremely Bright GRB 221009A.** J. Yang, X.-H. Zhao, et al., ApJL, doi: 10.3847/2041-8213/acc84b

- **GRB 220408B: A Three-Episode Burst from a Precessing Jet.** Z. Zhang, Y.-H. I. Yin, et al., RAA, doi: 10.1088/1674-4527/acfa59

2022 **A long-duration gamma-ray burst with a peculiar origin.** J. Yang, S. Ai, et al, Nature, doi: 10.1038/s41586-022-05403-8

2021 **GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions.** X. I. Wang, X. Zheng, et al., ApJ, doi: 10.3847/1538-4357/ac29bd

MEDIA COVERAGE

- **Triggering the Untriggered: The First Einstein Probe-Detected Gamma-Ray Burst 240219A and Its Implications** featured in **AAS Nova Journals Digest**, 31 March 2025.